P22 — Building Segmentation in Satellite Images

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1. Problem and question

We investigate binary building segmentation in aerial images: **does adding soft Dice to binary cross-entropy improve foreground overlap with the same U-Net + ResNet18 and nominal training settings?** The intended experimental variable is the loss function.

2. Dataset and preprocessing

Massachusetts Buildings, Kaggle repack version 2, contains 151 original 1500 × 1500 image–mask pairs. The original train/validation/test split has **137/4/10 images**, producing **4932/144/360 patches**. Pairs are zero-padded to 1536 × 1536 and divided into 36 non-overlapping 256 × 256 patches; all patches are retained. RGB inputs use ImageNet normalization; grayscale masks are binarized with `> 0`. Source filenames are disjoint across splits. Geographical/content duplication has not been excluded. **Dataset license confirmation remains pending.**

3. Method and experiment

Both conditions use U-Net with an ImageNet-pretrained ResNet18 encoder. The baseline uses `BCEWithLogitsLoss`; the comparison adds mean per-patch soft Dice loss, with equal coefficients and smoothing 1. Shared settings are Adam, learning rate 0.001, batch size 8, ten epochs, seed 42 and no augmentation. Checkpoints are selected by minimum validation loss under each condition's own objective.

The main comparison thresholds sigmoid probabilities at **0.5 for both models**. Metrics use global foreground pixel totals, including empty and padded regions: `IoU = I/(U + 1e-7)`; `Dice = 2I/(P + T + 1e-7)`. These are not per-patch averages.

4. Results and sensitivity

Model, common threshold 0.5	Val IoU	Val Dice	Test IoU	Test Dice
BCE baseline	0.6338	0.7759	0.6898	0.8164
BCE + soft Dice	0.6160	0.7624	0.6681	0.8010

Adding soft Dice reduced test IoU by **2.17 percentage points** and Dice by **1.54 percentage points** in this configuration.

Sensitivity analysis evaluated thresholds 0.3/0.5/0.7. Maximizing validation IoU selects **0.5 for BCE** (0.6338) and **0.3 for BCE + Dice** (0.6199). With these validation-selected thresholds, comparison test IoU/Dice are **0.6718/0.8037**; baseline remains **0.6898/0.8164**. Threshold 0.7 reduces validation overlap for both. This sensitivity result is separate from the common-threshold comparison.

5. Interpretation and limitations

The chosen Dice term did not improve overlap, even after threshold selection. This does not establish that Dice loss is generally inferior. One training run per condition, four validation source images, different GPUs (T4/RTX 5070), and incomplete baseline environment/execution-order records limit causal and statistical claims. BCE-only and combined-loss values are not directly comparable.

6. Failure analysis

Renata's results checkpoint report (pp. 2–3) and the failure comparison describe three test examples at threshold 0.5, regenerated during verification:

- **113, `23429020_15`, patch 05 — Observation:** a large roof is truncated at the source-image boundary. BCE recovers most of it (IoU 0.833); BCE + Dice misses most of it (0.085). **Possible cause:** limited spatial context may contribute to the omission. This is an untested hypothesis and does not establish why the models differ, especially since BCE succeeds on the same input.
- **216, `23579005_15`, patch 00 — Observation:** the labelled building occupies 553 pixels (0.844% of the patch); both models recover only part of it and produce false positives (IoU 0.147/0.172). **Possible cause:** its small spatial extent may make the complete shape difficult to recover at this image resolution.
- **267, `23729035_15`, patch 15 — Observation:** both models mark a bright object at the water's edge, while GT contains only 18 foreground pixels; IoU is zero. **Possible cause:** the false positives may reflect confusion between this object and building foreground. Its identity is uncertain; an annotation error has not been established.

These explanations are visually motivated hypotheses, not demonstrated causes.

7. Reproducibility and runtime

Repository: `https://github.com/1r444444/CV-P22-building-segmentation`. **Evaluation was repeated from both saved checkpoints on all 144 validation and 360 test patches; training was not rerun or independently reproduced.** Evaluation used macOS arm64, Python 3.11.15, PyTorch 2.11.0 and segmentation-models-pytorch 0.5.0. Baseline IoU/Dice at 0.5 match the saved values exactly. Comparison release caches reproduce recorded metrics exactly; fresh CPU inference agrees at four-decimal precision, with small numerical differences. Full evaluation and sensitivity analysis were executed sequentially in notebook 06.

Kristina estimates **about 18 minutes of training on T4** and **3–4 seconds of batched test inference over 360 patches**, batch size 8, 256 × 256 inputs, from progress bars. These are approximate historical estimates, not a separate benchmark or single-image latency. The comparison manifest records **312.19 s** for training including epoch validation/checkpoint writes and **1.49 s** for test evaluation including loading, forward passes, loss and metrics on RTX 5070. Different hardware and timing boundaries prevent a direct speed comparison.

8. Members' contributions

- **Kristina Soloveva:** dataset preparation, preprocessing and BCE baseline.
- **Elena Tuturina:** BCE + Dice implementation/run, artifacts and comparison milestone report.
- **Renata Salikhzyanova:** evaluation and failure analysis.
- **Irina Bychkova:** environment/integration fixes, reproducibility verification, technical summary, and development and verification of the fresh-inference demo script.

Recorded demo: Watch the pipeline demonstration (Google Drive).